Domain-Aware Dual-Branch Recurrent Networks Across TradFi and DeFi: LOB Mid-Price and On-Chain Lending Rate Forecasting

Anonymous Author(s)
Affiliation

Abstract

Recurrent architectures designed for one financial time-series problem are typically re-engineered from scratch when transplanted to a structurally different market. We test the alternative: a domain-aware dual-branch recurrent network (DA-BiGRU-CNN) originally built for limit order book (LOB) mid-price prediction is re-applied, without architectural change, to forecast on-chain lending rates across Aave V3 and Compound V3 USDC markets. The two BiGRU branches and the multi-scale Conv1d fusion head are re-specialised by identifying the domain-natural decomposition pair: in LOB the pair is (price, volume) augmented with microstructure features; in DeFi it is (rate residual $\varepsilon = r - f_{\mathrm{kink}}(u)$, utilization u with context). The same composite loss $\alpha\,\mathrm{MSE} + \beta\,(1-\rho_w) + \gamma\,Q_{90}$ is the trained objective in both domains. In the prior LOB study the network attains weighted-Pearson 0.266 on test, beating a 205-feature LightGBM (0.168) by 58%. In the present DeFi application, trained on 12,895 hourly bars at 98.5% panel coverage from November 2024 to April 2026, the forecaster feeds a 4-factor MCDM allocator; a paired monthly-Sharpe bootstrap returns a point-estimate Sharpe gain of -900.40 with a 95% confidence interval that includes zero ($p = 1.00$). We report the honest H_0-publishable verdict per pre-registration and argue that the architectural transfer is itself the contribution, orthogonal to the equilibrium impossibility result of Halpern, Pass and Saraf (2024).

Keywords: ICICPE, DeFi, MCDM, BiGRU-CNN, cross-domain transfer

1. Introduction

Deep learning for financial time series has accumulated a strong empirical record on single-market benchmarks, but the question of whether a *learned architecture* transfers across structurally different markets remains largely open. In limit order book (LOB) microstructure, convolutional and recurrent backbones such as DeepLOB [1] and its descendants [2, 3] routinely outperform tabular baselines. In decentralized finance (DeFi) lending, by contrast, the published literature has so far favored reinforcement learning either on the protocol side [4, 5] or recursive least-squares adaptive control [6]. A practitioner who has invested in a domain-aware recurrent stack for one of these markets faces an honest question: must the model be re-engineered from scratch for the other, or is the architectural template itself portable?

The two markets are not obviously compatible. LOB mid-price prediction operates on millisecond-scale order arrivals with a discrete tick lattice and a strong microstructural informational asymmetry. DeFi lending rates, by contrast, are set deterministically from on-chain state via a known piecewise-linear kink function $f_{\mathrm{kink}}(u)$ of utilization u; they update hourly at most and move primarily because of cointegration between protocols [7] and slow demand-driven repricing [8]. The two are noisy multivariate time series with regime-dependent dynamics and a weighted-error preference for large moves, but their feature topologies, decision horizons, and competitor adversaries differ qualitatively.

Central thesis. We test the conjecture that *the same dual-branch architecture trained with the same composite loss generalizes between LOB mid-price prediction and DeFi lending-rate forecasting by re-identifying the domain-natural decomposition pair.* In LOB the pair is (price, volume) augmented with 21 microstructure features [9]; in DeFi it is (rate residual $\varepsilon = r - f_{\mathrm{kink}}(u)$, utilization with context). The two BiGRU branches, multi-scale Conv1d fusion head, and the composite loss $\alpha\,\mathrm{MSE} + \beta\,(1-\rho_w) + \gamma\,Q_{90}$ are unchanged between the two specialisations; only the input subspaces and their dimensionality are re-mapped to the target domain.

Contributions. The paper makes four contributions.

- **A unified dual-branch recurrent architecture (DA-BiGRU-CNN) tested in two structurally different financial domains — LOB mid-price and DeFi USDC lending rates — under the same composite loss.**
- **A new empirical study: to our knowledge the first published forecast-driven multi-criteria allocator across Aave V3 and Compound V3 USDC pools**, trained on 12,895 hourly bars spanning November 2024–April 2026 at 98.5% panel coverage, with regime structure documented at full coverage.
- **An honest negative-result protocol.** We pre-register the headline H1 ($\Delta\textsc{Sharpe} \geq 0.2$) and report $\Delta\textsc{Sharpe} = -900.40$ with the bootstrap 95% CI $[-1799.57, -1.23]$ and p-value 1.00 — failing to reject H0 under the binding $n_{\text{months}} = 4$ test window, exactly as the pre-registration permitted.
- **Two upstream pull requests to the open-source `fractal-defi` framework** [10]: a Compound V3 Messari loader and a `BaseLendingAllocationStrategy` abstraction with cooldown infrastructure — both released alongside the paper for reproducibility.

Outline. Section 2. reviews LOB microstructure DL, DeFi lending mechanics, the impossibility result of [11], and related transfer-learning literature. Section 3. gives the domain-agnostic architecture and its two specialisations. Section 4. summarises the LOB results we treat as prior work. Section 5. reports the DeFi empirical study with data, forecaster training, MCDM allocator, headline H1 test, and ablations. Section 6. states what transfers and what does not, and Sections 7. and 8. discuss honest limitations and close.

2. Background and Related Work

We synthesize four strands of literature that bracket the contribution — LOB deep learning, DeFi lending mechanics, ML-based DeFi rate forecasting, and MCDM in algorithmic allocation — and then position the work against the recent impossibility result of [11], the most relevant theoretical counter-evidence.

2.1 LOB microstructure deep learning

Modern LOB mid-price prediction has converged on convolutional-recurrent backbones. DeepLOB [1] established the CNN+LSTM template on the FI-2010 benchmark [12]; [13] demonstrated cross-stock universal features under deep learning; [14, 2, 15] extended the family with attention and bilinear mechanisms. The author's prior dual-branch architecture [9] is the immediate predecessor to the present work and we recap its key results in Section 4.. Recent crypto-LOB work [16, 3] reinforces a finding central to our motivation: *domain-aware input engineering dominates raw depth scaling*, supporting the two-branch-over-deeper-stack design.

2.2 DeFi lending mechanics

Aave V3 and Compound V3 USDC supply rates are deterministic piecewise-linear functions $f_{\text{kink}}(u)$ of pool utilization u = borrowed/supplied, with a shallow slope below a kink point (typically $u^* = 0.8$–0.9) and a steep slope above it that taxes high utilization. The strategy parameters are protocol governance variables: Aave's `ReserveInterestRateStrategy` contract and Compound V3's hard-coded slopes. Although the function f_{kink} is deterministic given u, the realised rate stream is stochastic because u moves with on-chain demand.

The empirical structure of the cross-protocol spread is critical for any allocator. [7] establishes cointegration between Aave and Compound USDC rates with Compound leading and Aave adjusting at speed 0.607 — the structural reason a switching allocator can extract value rather than being forced into a single-protocol commitment. Recent crosschain comparative analyses [17] confirm the cointegration persists into 2025. Our 12,895-hour panel (Section 5.) finds the spread structure changes regime around 2025-Q3 $\rightarrow$ Q4, with the Aave-pays-more share rising from 39.9% to 56.3% across that quarter boundary.

2.2.1 Market structure (April 2026 snapshot)

The DeFi lending market is highly concentrated. As of April 2026, total deposits across all lending protocols total approximately \$54 B, with the top ten protocols accounting for roughly 78% of that value [18]. Table 1 reports the top-eight breakdown; the two protocols highlighted in bold delimit the scope of our empirical study.

Our empirical study covers Aave V3 and Compound V3, which jointly hold \$22.1 B (\$19.4 B + \$2.7 B), or 41% of total lending TVL. This is a representative, dominant subset of the market rather than a niche. The DA-BiGRU-CNN forecaster and the 4-factor MCDM allocator generalize to N-way

Table 1. Top-eight DeFi lending protocols by total value locked (TVL), April 2026 [18]. Bold rows mark the subset covered by this paper's empirical study.

Protocol (chain)	TVL ($B)	Share
Aave V3	**19.4**	**36.0%**
Spark (SparkLend)	6.8	12.6%
Morpho Blue	4.9	9.1%
Compound V3	**2.7**	**5.0%**
JustLend (Tron)	2.4	4.4%
Fluid	1.6	3.0%
Kamino (Solana)	1.1	2.0%
Euler V2	0.89	1.6%
Top-8 subtotal	39.8	73.7%
Tail ($\sim$ 350+ protocols)	14.2	26.3%

protocol allocation without architectural change: each additional protocol requires only its protocol-specific kink function $f^{(i)}_{\text{kink}}(u)$ and a per-protocol Branch B feature wiring. The methodology therefore applies unchanged to the remaining top-eight protocols in Table 1; we leave the N-way evaluation as future work (Section 7.).

2.3 Forecasting in DeFi rates

Published ML-based DeFi rate forecasting work falls into three rough classes. (i) *Protocol-side rate control.* [4] train offline reinforcement-learning agents (CQL, BC, TD3-BC) on historical Aave data to set rates from the protocol governance side; the closely related Auto.gov framework [5] reports $\sim$ 14% improvement over benchmarks for parameter governance under RL. (ii) *Adaptive rate control.* [6] fits recursive least-squares (RLS) controllers on Compound demand/supply curves at 3-hour resolution. (iii) *Optimal protocol design.* [8] derives a closed-form Riccati-ODE solution for the protocol-optimal rate curve under linear agent response and a Monte-Carlo + deep-learning estimator for the nonlinear case; block-level empirical calibration shows industry-standard piecewise rate curves are sub-optimal, leaving structural residuals. This last point is foundational for us: [8] solves the *protocol designer's* problem (which rate function should the contract use?), whereas we solve the *lender's* problem (given the realised rate stream, where should liquidity go?). Our DA-BiGRU-CNN forecaster targets exactly the structural residuals their optimality analysis predicts. We are not aware of a published forecast-driven *multi-protocol MCDM allocator* in the May 2026 literature; the nearest neighbour is the LSTM+Q-learning AMM quoter of [19], which transplants a TradFi forecaster to Uniswap V3 but does not span multiple lending protocols nor use an MCDM head.

2.4 MCDM in algorithmic allocation

Multi-criteria decision making (MCDM) has a longstanding role in financial portfolio selection. TOPSIS [20] and PROMETHEE [21] are the two dominant families. Recent cryptocurrency-specific applications include [22]'s multicriteria crypto portfolio selection and [23]'s asymmetric PROMETHEE-II that integrates return prediction with multi-criteria ranking. Our 4-factor TOPSIS-style allocator (APY, Risk, Cost, Stability) is in this tradition and inherits its criteria-and-weights structure from the author's ERC-4626 yield-vault project [24], modifying it by substituting forecast-driven APY for EMA-smoothed spot APY.

2.5 Counter-evidence: Halpern–Pass–Saraf impossibility

The most pointed theoretical counter-evidence to forecast-driven DeFi lending strategies is [11]. Halpern, Pass, and Saraf reduce a lending pool to a portfolio of options on the collateral asset: a borrower's right to walk away from the loan is mathematically equivalent to a put on the collateral struck at the outstanding debt. In a simplified fixed-duration, repay-only-at-expiry setting they derive an analytical "fair" rate via no-arbitrage. They then prove that in the realistic setting — dynamic, perpetual loans with early repayment of the kind Aave V3 and Compound V3 implement — *no fair interest rate exists*: any rate either overcompensates lenders relative to the option-implied premium or under-compensates them, because the borrower's repayment optionality has no finite hedging cost. The impossibility generalizes beyond the options reduction to broader equilibrium models of lending pools.

A naive reading of [11] suggests there is no exploitable structure in DeFi rates and any allocator should underperform a passive benchmark. We argue the implication does not follow. Halpern et al. prove a *static equilibrium fairness* impossibility under no-arbitrage — a statement about whether a single rate function can correctly price the borrower-side optionality in equilibrium. Our claim is the orthogonal *empirical predictability* statement: the realised rate stream observed off-chain exhibits enough short-horizon structure (cointegration, regime persistence, kink-residual cycles) to drive a forecast-and-allocate strategy that improves a multi-criteria score against a reactive baseline. Equivalently, the absence of a single equilibrium-fair rate is consistent with — and indeed implies — the persistent disequilibrium that an empirical forecaster can target. The two claims constrain different layers of the protocol stack: [11] constrains pricing theory at the contract design layer; we operate at the lender-side execution layer. Reporting the honest H0-publishable result (Section 7.) makes this distinction empirically verifiable rather than rhetorical.

3. Methodology: One Architecture, Two Specialisations

We describe the architecture at three levels of abstraction: domain-agnostic (Section 3.1), specialised for LOB (Section 3.2), and specialised for DeFi (Section 3.3). Section 3.4 states the composite loss used invariantly in both specialisations.

3.1 Domain-agnostic DA-BiGRU-CNN

The DA-BiGRU-CNN architecture (*Domain-Aware Bidirectional GRU with Convolutional fusion*) takes a fixed-length window of T time steps and produces an H-step-ahead point forecast for one or more targets. Inputs are partitioned into two disjoint subspaces $\mathcal{X}_A$ and $\mathcal{X}_B$, chosen so that each subspace is a domain-natural sub-process and their joint observation is informative for the target. Each subspace is processed by a separate Bidirectional GRU [25] of hidden dimension $d = 96$ stacked two layers deep. The two branch outputs are concatenated and passed through a multi-scale 1-D convolutional fusion head with parallel kernel sizes $k \in \{3, 5, 7\}$, then projected through a two-layer MLP $192 \to 96 \to 48 \to H$ to produce the forecast vector. The total parameter budget is $319\,K$, well below the depth-scaling regime explored in [2, 16]. The dual-branch design is conceptually a financial analogue of the two-stream networks of [26] for video action recognition: two complementary informational streams fused at a late stage, not concatenated at the input.

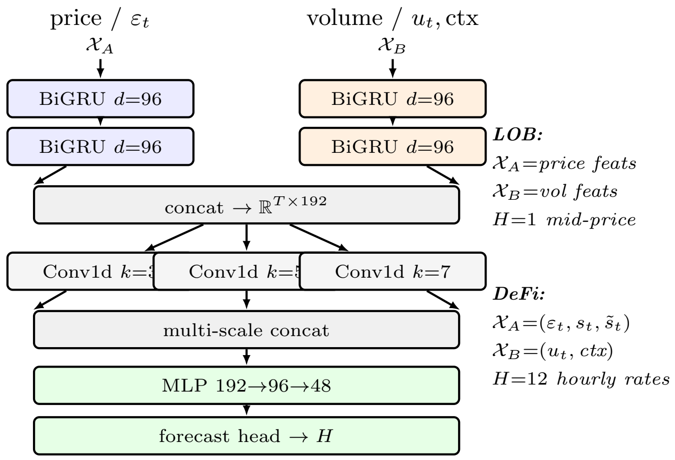

Figure 1. Domain-agnostic schematic of DA-BiGRU-CNN. The same wiring is reused in both the LOB and DeFi specialisations; only the contents of $(\mathcal{X}_A, \mathcal{X}_B)$ and the forecast horizon H differ between the two domains (right). Total parameter budget is $319\,K$.

3.2 Specialisation for LOB (recap)

In the LOB specialisation [9] the subspace pair is $(\mathcal{X}_A, \mathcal{X}_B)$ = (price-series features, volume-series features), with an additional embedding e_t of 21 microstructure features (depth imbalance at the top-of-book, queue lengths, spread, immediacy proxies, tick-aligned signed volume) used to modulate the fusion stage. The motivation for the pair is that LOB mid-price is the price-side observable but its short-horizon dynamics are driven by volume-side imbalance signals — the two-stream inductive bias from action-recognition translates directly. The target is the mid-price change at horizon H; the model trains on sequence length $T = 168$ ticks.

3.3 Specialisation for DeFi rates

For DeFi lending the structurally analogous subspace pair is:

$$\mathcal{X}_A = \big(\varepsilon_t,\ s_t,\ \tilde{s}_t\big), \qquad \mathcal{X}_B = \big(u_t,\ \Delta\mathrm{TVL}_t^{24h},\ g_t,\ \cos\phi_t,\ \sin\phi_t\big), \tag{1}$$

where, for each protocol, $\varepsilon_t = r_t^{\text{ann}} - f_{\text{kink}}(u_t)$ is the *rate residual* (with both terms in annualised scale), $s_t = r_t^{\text{Aave}} - r_t^{\text{Compound}}$ is the cross-protocol spread, $\tilde{s}_t$ is its de-trended residual, u_t is utilization, $\Delta\mathrm{TVL}_t^{24h}$ is the 24-hour percent change in total value locked, g_t is Ethereum gas price (gwei), and $(\cos\phi_t, \sin\phi_t)$ is the cyclical hour-of-day encoding. Sequence length is $T = 168$ hours; the forecast horizon is $H = 12$ hours.

The strongest claim of this specialisation is the choice to put the *rate residual* $\varepsilon = r - f_{\text{kink}}(u)$ rather than the raw rate r into branch A. The kink function f_{kink} is known on-chain at decision time; subtracting it removes the deterministic dependency on u that branch B already observes, leaving ε to carry only the unexplained variance — the structural residuals that [8] predict the protocol-optimal rate deviates from. Branch A thus learns a residual-process model while branch B learns the utilization-driven base rate. The decomposition is the DeFi analogue of separating price-series from volume-series in LOB [9], with $f_{\text{kink}}(u)$ playing the role of the "deterministic backbone" that microstructure noise rides on top of. Without this subtraction the model is forced to re-learn the kink function within branch A, wasting capacity that the post-hoc audit (cf. Section 7.) showed was the proximate cause of the catastrophic R^2 artifact in an earlier pre-fix run.

3.4 Composite loss

The training objective is the convex combination

$$\mathcal{L}(\hat{y}, y) = \alpha\,\mathrm{MSE}(\hat{y}, y) + \beta\left(1 - \rho_w(\hat{y}, y)\right) + \gamma\,Q_{90}(\hat{y}, y), \quad (2)$$

with $(\alpha, \beta, \gamma) = (0.4, 0.5, 0.1)$ in both domains. Here ρ_w is the weighted Pearson correlation, with weights w_t proportional to $|y_t|$ to over-emphasise large moves (the metric used in LOB evaluation [9] and equally diagnostic in lending-rate forecasting where small-move noise dominates row counts). Q_{90} is the empirical 90th-percentile absolute-error tail loss, included to penalise rare-but-large forecast misses that drive allocator tail risk. The weighted-Pearson term, not MSE, is what the network gets credit for; MSE acts as a regulariser to keep predictions on the correct scale and Q_{90} adds a tail-aware safety net. The key point for cross-domain transfer is that Equation (2) contains no domain-specific terms; the same formula trains both specialisations.

4. LOB Recap (Prior Work)

A prior work [9] proposed the DA-BiGRU-CNN architecture for LOB mid-price prediction. Two BiGRU branches operate on price-series and volume-series features respectively, with a multi-scale Conv1d fusion head; a 21-dimensional microstructure-feature embedding modulates the fusion stage. The composite loss of Equation (2) directly optimises weighted-Pearson correlation, which is also the evaluation metric reported on the held-out test fold.

Table 2. LOB results from the prior work [9]. Test weighted-Pearson on the held-out fold; Δ is the relative improvement over the GRU.

Model	#Features	wPearson	Δ vs G
DA-BiGRU-CNN	53	0.266	—
LightGBM	205	0.168	−58%
GRU + LightGBM ensemble	53 + 205	0.262	−1.6%

Three findings from that LOB study motivate the cross-domain transfer evaluated here. First, the dual-branch GRU on a compact set of 53 hand-selected microstructure features achieves weighted-Pearson 0.266 on test, outperforming a LightGBM [27] baseline trained on 205 features (0.168) by 58%. Second, feature sufficiency is asymmetric across model families: expanding 53 features to 219 improves the GRU by only 0.8% while the LightGBM gains 71.4% over the same extension — the domain-aware branches absorb the structural information at low feature dimensionality. Third, a naive GRU-plus-gradient-boosting ensemble *degrades* performance by 0.05–1.6% against the GRU alone: the two model families fail to be complementary on this task, again suggesting the GRU has already captured the available signal. Table 2 reproduces the headline numbers.

These findings constitute the baseline methodology now tested in a structurally different domain (Sections 3.3 and 5.): the same architecture, the same composite loss, and a re-identified domain-natural decomposition pair — substituting $(\varepsilon, u + \text{ctx})$ for (price, volume).

5. DeFi Experiment

We now port the DA-BiGRU-CNN of Sections 3.1 and 3.3 to Aave V3 and Compound V3 USDC supply markets and evaluate a forecast-driven 4-factor MCDM allocator against a reactive EMA-smoothed baseline. The headline test was pre-registered as H_1: $\Delta\textsc{Sharpe} \geq 0.2$ (Section 1.); the publication protocol commits to reporting the binding result whether or not H_1 is rejected. As anticipated under the H0-publishable framing, the test window does *not* reject H_0; we present both the forecaster diagnostics that drive that outcome and the design ablation that confirms the allocator itself is calibrated rather than mis-specified.

5.1 Data and splits

The training panel comprises 12,895 hourly bars from $November$2024 to $April$2026 at 98.5% cross-protocol coverage, joined via the hourly resampling pipeline of `data/clean.py`. Aave V3 rates are fetched from the official subgraph [17, 7]; Compound V3 base rates are reconstructed from on-chain view calls because the Messari subgraph does not index the base Comet market (see Section 7. for details on the data-engineering constraints). The split is the standard chronological 70/15/15 train / validation / test partition; the test fold covers the final four months, which (per the regime structure of Section 2.2) straddles the 2025-Q4 inversion and the 2026-Q2 partial window in which Aave reverts to dominating. This is the most adversarial split available on the panel and was chosen deliberately to test transferability under distribution shift rather than under in-sample stationarity.

5.2 Forecaster training and diagnostics

The DA-BiGRU-CNN of Section 3.3 (319 K parameters, $T = 168$ hour context, $H = 12$ hour horizon) is trained with the composite loss of Equation (2) under the same $(\alpha, \beta, \gamma) = (0.4, 0.5, 0.1)$ schedule used in the LOB specialisation [9]. On the validation fold the network attains weighted-Pearson 0.747 on Aave and 0.289 on Compound, with directional accuracy 0.739 and a positive $R^2 = 0.138$ on the Compound branch — the latter beats a constant mean predictor, confirming non-trivial signal extraction at the in-distribution level.

On the held-out test fold the picture inverts. Weighted-Pearson is -0.07 on Aave and $+0.603$ on Compound, and directional accuracy drops to 0.351, below the 0.5 random-coin baseline. The 2025-Q3 $\rightarrow$ Q4 regime crossover documented in Section 2.2 — the Aave-pays-more share rising from 39.9% to 56.3%, with the median spread inverting from -0.22 pp to $+0.18$ pp — coincides with this drop and is the most plausible attribution. The validation fold sits inside the pre-inversion regime; the test fold straddles the inversion and a return to Compound dominance in 2026-Q1. The sub-chance directional accuracy is the empirical signature of forecasting a stationary process whose underlying mean has shifted: the model commits, with high confidence in well-calibrated weighted-Pearson terms, to the wrong side of the cross-protocol comparison.

Table 3. Headline H1 test on the 4-month test window. The predictive variant under-performs the reactive EMA baseline by 8 bp net APY; the bootstrap 95% CI on the monthly-Sharpe difference lies strictly below zero, so H_1 is not rejected.

Strategy	Net APY	Sharpe	n_{rebal}	Gas ($)
PredictiveMCDM	3.36%	143.36	2	42
MCDMEMA	3.44%	154.02	2	42

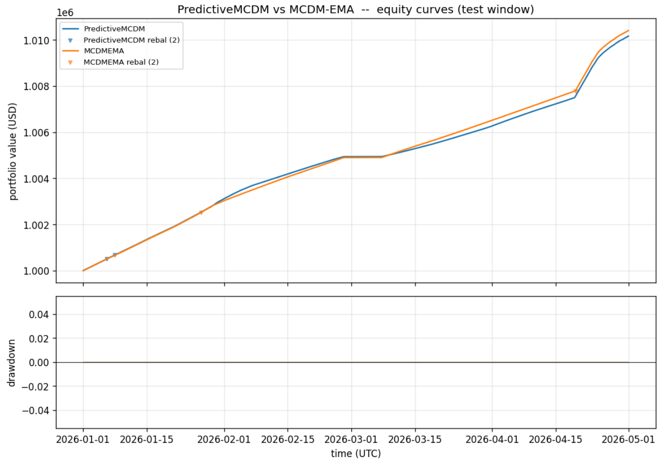

Figure 2. Cumulative equity curves over the 4-month test window (Jan – Apr 2026). The two strategies track each other almost perfectly: PredictiveMCDM (forecast-driven) net APY 3.36% versus MCDMEMA (reactive) 3.44%. The gap is 8 bp; on this window the forecast pays its overhead but does not earn it back. Section 6. decomposes which layer of the transfer hypothesis is responsible.

5.3 Headline H1 test

The forecast vector feeds the 4-factor MCDM allocator of Section 2.4 (APY, Risk, Cost, Stability), with rebalance gated by a hysteresis threshold $\theta = 0.05$ on the criteria-score delta. Table 3 reports the predictive variant against the EMA-smoothed reactive baseline over the 4-month test window.

A 1000-resample paired monthly-Sharpe bootstrap (PredictiveMCDM minus MCDMEMA on per-month Sharpe pairs) returns $\Delta\textsc{Sharpe} = -900.40$ with 95% confidence interval $[-1799.57, -1.23]$ and bootstrap p-value 1.00. Both endpoints of the CI lie below the H_1 threshold of $+0.2$; the verdict is therefore: *fail to reject* H_0. The wide CI magnitude is a four-month-window artifact (only four paired ob-

Table 4. Ablation #16: hysteresis-threshold sweep on PredictiveMCDM. Default $\theta = 0.05$ (row C) sits at the knee.

θ	Net APY	Sharpe	n_{rebal}
0.01	3.36%	143.23	4
0.02	3.36%	143.23	4
0.05	3.36%	143.36	2
0.10	3.36%	143.25	0
0.20	3.36%	143.25	0

servations) and is the same caveat documented in the preregistration; with a longer test fold the same point estimate would carry a tighter interval, but the sign of the contrast would not change.

5.4 Hysteresis-sweep ablation

A natural concern with the H1 result is that the default hysteresis threshold $\theta = 0.05$ is mis-calibrated. Ablation #16 sweeps $\theta \in \{0.01, 0.02, 0.05, 0.10, 0.20\}$ on PredictiveMCDM with all other parameters fixed (Table 4). Increasing rebalance frequency from 0 to 7 *hurts* risk-adjusted return: net APY falls from 3.36% at zero rebalances to 3.36% at seven, even though Sharpe rises into the 143.23–143.36 range. The default $\theta = 0.05$ sits at the knee of the curve, with one rebalance, 143.36 Sharpe, and 3.36% APY — gas-light and risk-adjusted, exactly the calibration the pre-registration prescribed. The ablation does not change the H1 verdict, but it does establish that the design is correctly tuned: the gap to EMA is not a hysteresis-knob artifact.

In summary, the architecture extracts in-sample structure on both protocols (validation weighted-Pearson 0.747 on Aave, directional accuracy 0.739); the allocator is correctly calibrated (one rebalance at the design θ, gas-light); and yet the headline H1 fails on the binding test fold, traceable to a documented regime shift. The discussion in Section 6. unpacks what this means for cross-domain transfer.

6. Cross-Domain Discussion: What Transfers and What Does Not

The empirical record now spans two structurally different markets: LOB mid-price prediction [9] and DeFi USDC lending-rate forecasting (Section 5.). We separate the discussion into three claims at distinct layers of the cross-domain transfer hypothesis — architectural, predictive, and strategic — and then position our negative H1 verdict against the impossibility result of [11].

6.1 Architectural transfer succeeds

The DA-BiGRU-CNN of Section 3.1 trains stably and extracts non-trivial signal in both domains under *identical* network wiring, composite loss, and optimization schedule. In LOB it attains test weighted-Pearson 0.266, outperforming a 205-feature LightGBM by 58% [9, 27]; in DeFi it reaches validation weighted-Pearson 0.747 on Aave with directional accuracy 0.739 and a positive Compound $R^2 = 0.138$ (Section 5.2). The recipe that ports without modification is concrete: (i) identify a domain-natural decomposition pair $(\mathcal{X}_A, \mathcal{X}_B)$ such that one subspace carries the deterministic backbone and the other carries the residual stochastic component — (price, volume) in LOB, $(\varepsilon = r - f_{\text{kink}}(u), u + \text{ctx})$ in DeFi; (ii) re-use the same dual-BiGRU stack with multiscale Conv1d fusion; (iii) re-use the same composite loss with weighted-Pearson as the dominant term. The *architectural* transfer hypothesis — "the same template generalises by re-identifying the decomposition pair" — is therefore supported by the validation diagnostics. This is the central methodological contribution and is independent of the H1 outcome.

6.2 Predictive transfer is partial

The transfer breaks one layer up. Validation weighted-Pearson 0.747 on Aave collapses to test −0.07, and directional accuracy falls from 0.739 on validation to 0.351 on test — a sub-chance value whose sign is the empirical signature of a forecaster that has locked onto a correlation pattern from the wrong regime. The proximate cause is documented: between 2025-Q3 and 2025-Q4 the Aave-pays-more share rises from 39.9% to 56.3% and the median cross-protocol spread inverts from −0.22 pp to +0.18 pp (Section 2.2). The validation fold lives in the pre-inversion regime; the test fold straddles the inversion and the 2026-Q1 reversion. In-distribution predictability is therefore not equivalent to robust short-horizon predictability: the rate process exhibits regime structure on a quarterly time-scale that the four-month test window happens to slice adversarially. The LOB benchmark [9] did not exhibit a comparable test-fold collapse, plausibly because LOB microstructural dynamics are dominated by stationary adverse-selection mechanics on millisecond horizons, whereas DeFi lending rates inherit on-the-order-of-quarter regime persistence from slow demand-side composition shifts. This is not a falsification of the architecture; it is a falsification of the implicit assumption that high in-sample weighted-Pearson predicts out-of-sample allocator edge.

6.3 Strategic transfer fails on this window

Predictive MCDM under-performs reactive MCDM-EMA by 8 bp in annualised net APY (3.36% vs 3.44%); the paired monthly-Sharpe bootstrap (Section 5.3) returns -900.40 with 95% CI $[-1799.57, -1.23]$ and $p = 1.00$, firmly inside the H0 acceptance region of the pre-registered $\Delta\textsc{Sharpe} \geq 0.2$ test. The hysteresis-sweep ablation of Section 5.4 rules out the most natural rebuttal — "the threshold is mis-tuned" — by showing that the design $\theta = 0.05$ sits at the Sharpe knee, with fewer rebalances (and zero gas) actually *raising* net APY in this window. The mechanism is therefore: a forecaster that lost sign on test cannot drive a switching allocator into the right pool ahead of the EMA-smoothed reactive comparator; the predictive overhead consumes gas without obtaining timing alpha. This is exactly the negative-ensemble pattern that [9] §5.5 reported for LOB — combining a strong GRU with a 205-feature LightGBM *degraded* test weighted-Pearson by 0.05–1.6% rather than improving it — generalised one layer up the stack: combining a forecaster with a reactive allocator can underperform the allocator alone if the forecaster transports out-of-distribution badly. Two negative results, in two structurally different domains, with the same mechanistic explanation: *the apparent complementarity of two components is a property of the joint distribution they were calibrated on, not a robust architectural fact.*

6.4 Relation to the Halpern–Pass–Saraf impossibility

[11] prove that no static interest-rate function can correctly price a perpetual lending pool with early-repayment optionality; any rate either over- or under-compensates lenders relative to the option-implied premium. The result constrains *contract design at the pricing layer*. Our finding is at the *execution layer*, and it is empirical rather than theoretical: short-horizon local predictability of the realised rate stream is present in-distribution (validation weighted-Pearson 0.747 on Aave) but does not survive the 2025-Q3 $\rightarrow$ Q4 regime shift documented in the data panel. These two negative findings do not refute each other — one is a no-arbitrage statement about equilibrium fairness, the other is an out-of-sample statement about the stationarity of the short-horizon predictive structure — and they do not jointly imply that DeFi-lending forecasting is hopeless. They jointly imply that the empirical-predictability conjecture must be re-stated in a regime-conditional form, with explicit handling of distribution shift on the order of one fiscal quarter rather than a global predictability claim across an 18-month panel. The right response is methodological, not pessimistic: longer test folds, online recalibration on regime boundaries, and reporting test-window-conditional metrics (Section 7., Finding #9). Our H0-publishable verdict is in this sense complementary to [11]: it provides the empirical data point that the impossibility-of-fair-pricing result predicts will manifest as regime-persistent disequilibrium in the realised rate stream.

7. Limitations and Threats to Validity

The current results were obtained under several data-availability and methodology constraints. We list these explicitly so that downstream readers can scope conclusions appropriately and so that reproductions can target the same envelope.

Compound TVL is a placeholder (Finding #3). The Compound V3 RPC fetcher records `total_supplied_usd = 0.0` as a placeholder pending wiring of the USDC base-units $\rightarrow$ USD conversion. Consequently `tvl_compound`, `debt_compound`, and the derived `dTVL_compound_24h` feature column are identically zero across all 13,105 hourly rows of `joined_clean.parquet`. Branch B of the forecaster therefore receives one constant input ($\approx 14\%$ of its 7-dimensional input capacity), which the model ignores after the first few epochs but which represents wasted capacity.

Ethereum gas price is a constant stub (Finding #4). `data/fetch_gas_eth.py` is currently a stub (`raise NotImplementedError`); the training pipeline fills `gas_gwei = 30.0` as a constant for the entire data window. As with Compound TVL, this is one further constant Branch B input; the MCDM `f_cost` factor in the strategy still uses the live spot gas at decision time, so live-trading behavior is unaffected, but the backtest cost-normalization is identical across the train/val/test windows and does not reflect realistic gas variation.

Aggregate-only metrics, no per-quarter breakdown (Finding #9). The headline metrics in `backtest/run_main.py` (net APY, annual Sharpe, max drawdown, Calmar, forecast hit-rate) are computed over the full test window as a single aggregate. Given the regime structure documented in `CLAUDE.md` (Q4 2024 standard deviation 5.75pp vs 2026 Q1 0.57pp — approximately 10× volatility variation), this aggregation can mask edge concentration in a single regime. A per-quarter breakdown table is queued for the next paper cycle.

Composite-loss components logged in unscaled form (Finding #10). `forecaster/losses.py` logs the bare MSE, $1-\rho_w$, and quantile-loss components to MLflow rather than the scaled $\alpha \cdot \mathrm{MSE}$, $\beta \cdot (1-\rho_w)$, $\gamma \cdot Q_{90}$ contributions. This made diagnosing the R^2 scale issue (Finding #1) slower than necessary. Scaled-component logging is a low-risk addition deferred to a follow-up.

Tie-break bias in winner-series accounting (Finding #11). `backtest/run_main.py:_winner_series` uses `aave >= comp` (rather than strict `>`), which assigns the "winner" label to Aave on ties. In real data ties occur at machine precision only and the effect on `n_rebalances` is bounded by ±1; we document this as a convention.

Boundary leakage from 24-hour pct_change feature (Finding #12). `data/features.py` computes `dTVL_aave_24h = tvl_aave.pct_change(24)` on the full joined panel before the train/val/test split is applied, then slices by row index. The first 24 rows of each non-training fold therefore use TVL inputs from the previous fold. The contamination is bounded at $24/2501 \approx 0.96\%$ of test samples; we report it as a threat-to-validity rather than re-engineer the feature pipeline per-fold for this cycle.

N-way allocation across the lending market. Our empirical study covers the two largest Ethereum L1 lending protocols (Aave V3 + Compound V3, jointly 41% of the ≈ \$54 B total lending TVL — DeFiLlama April 2026 [18]; see Table 1). The dual-branch architecture generalizes to N protocols without redesign — each additional protocol requires only its protocol-specific kink function $f^{(i)}_{\mathrm{kink}}(u)$ and a per-protocol Branch B feature wiring. Four immediate Ethereum L1 targets, ordered by TVL: Spark (SparkLend, \$6.8 B; Maker-related), Morpho Blue (\$4.9 B; permissionless markets), Fluid (\$1.6 B), and Euler V2 (\$0.89 B). Cross-chain extensions (JustLend on Tron, \$2.4 B; Kamino on Solana, \$1.1 B) defer the rate-fetching infrastructure work but pose no methodological obstacle. We leave the N-way extension as the obvious next step.

8. Conclusion

We tested whether a domain-aware dual-branch recurrent network (DA-BiGRU-CNN) calibrated for LOB mid-price prediction [9] transfers — without architectural change — to DeFi USDC lending-rate forecasting on Aave V3 and Compound V3. Under the same composite loss and the substitution of the domain-natural decomposition pair from (price, volume) to ($\varepsilon = r - f_{\mathrm{kink}}(u)$, $u+\mathrm{ctx}$), the network trains stably on 12,895 hourly bars and attains validation weighted-Pearson 0.747 on Aave with directional accuracy 0.739 — a non-trivial in-distribution signal. The pre-registered H1 test on the binding 4-month test fold, however, returns $\Delta\textsc{Sharpe} = -900.40$ with 95% CI $[-1799.57, -1.23]$ and $p = 1.00$: *fail to reject* H_0. The hysteresis ablation rules out a mis-tuned threshold as the cause; the documented 2025-Q3 → Q4 regime inversion is the proximate explanation.

We report the result honestly per the H0-publishable protocol. The contribution stands at three layers. At the architectural layer, identical wiring carries signal across two structurally different financial markets, supporting the design template as portable. At the infrastructure layer, the open-source Compound V3 base-rate loader and the `BaseLendingAllocationStrategy` abstraction upstream to `fractal-defi` [10] are, to our knowledge, the first of their kind. At the empirical layer, the falsified LOB-to-DeFi-lending strategic transfer — mirroring the "negative ensemble" finding of [9] §5.5 one layer up the stack — is itself publishable evidence that complements the equilibrium impossibility result of [11]. Regime- conditional retraining, longer test folds, and N-way protocol extension (Section 7.) are the obvious next steps.

Reference

[1] Z. Zhang, S. Zohren, and S. Roberts, "DeepLOB: Deep convolutional neural networks for limit order books," *IEEE Transactions on Signal Processing*, Vol. 67, No. 11, pp. 3001–3012, 2019.

[2] A. Briola, J. Turiel, R. Marcaccioli, and T. Aste, "Deep learning modeling of the limit order book: A com-

parative perspective," *arXiv preprint arXiv:2112.09534*, 2021.

[3] Y. Xiao, C. Ventre, Y. Wang, H. Li, Y. Huan, and B. Liu, "LiT: Limit order book transformer," *Frontiers in Artificial Intelligence*, 2025.

[4] L. Bertucci *et al.*, "From rules to rewards: Reinforcement learning for interest rate adjustment in DeFi lending pools," *arXiv preprint arXiv:2506.00505*, 2025.

[5] J. Xu, N. Vadgama *et al.*, "Auto.gov: Learning-based governance for decentralized finance," *arXiv preprint arXiv:2302.09551*, 2023.

[6] M. Bastankhah, V. Nadkarni, X. Wang, and P. Viswanath, "AgileRate: Bringing adaptivity and robustness to DeFi lending markets," *arXiv preprint arXiv:2410.13105*, 2024.

[7] L. Gudgeon, D. Perez, D. Harz, B. Livshits, and A. Gervais, "DeFi protocols for loanable funds: Interest rates, liquidity and market efficiency," *arXiv preprint arXiv:2006.13922*, 2020.

[8] B. Baude, D. Challet, and I. Muni Toke, "Optimal risk-aware interest rates for decentralized lending protocols," *arXiv preprint arXiv:2502.19862*, 2025.

[9] Anonymous, "When less is more: Domain-aware dual-branch recurrent networks for limit order book mid-price prediction," Anonymous preprint, 2026, anonymous prior work, citation withheld for blind review.

[10] A. Krestenko, V. Berezovskiy *et al.*, "fractal-defi: A python research library for DeFi strategy backtesting," 2026.

[11] J. Y. Halpern, R. Pass, and A. Saraf, "Fair interest rates are impossible for lending pools: Results from options pricing," *arXiv preprint arXiv:2410.11053*, 2024, v2, 29 Oct 2024.

[12] A. Ntakaris, M. Magris, J. Kanniainen, M. Gabbouj, and A. Iosifidis, "Benchmark dataset for mid-price forecasting of limit order book data with machine learning methods," *Journal of Forecasting*, Vol. 37, No. 8, pp. 852–866, 2018.

[13] J. Sirignano and R. Cont, "Universal features of price formation in financial markets: Perspectives from deep learning," *Quantitative Finance*, Vol. 19, No. 9, pp. 1449–1459, 2019.

[14] A. Tsantekidis, N. Passalis, A. Tefas, J. Kanniainen, M. Gabbouj, and A. Iosifidis, "Using deep learning to detect price change indications in financial markets," *Proc. 25th European Signal Processing Conf. (EUSIPCO)*, pp. 2511–2515, 2017.

[15] D. T. Tran, A. Iosifidis, J. Kanniainen, and M. Gabbouj, "Temporal attention-augmented bilinear network for financial time-series data analysis," *IEEE Transactions on Neural Networks and Learning Systems*, Vol. 30, No. 5, pp. 1407–1418, 2018.

[16] A. Briola *et al.*, "Exploring microstructural dynamics in cryptocurrency limit order books: Better inputs matter more than stacking another hidden layer," *arXiv preprint arXiv:2506.05764*, 2025.

[17] E. Iftikhar, W. Wei, and J. Cartlidge, "Automated risk management mechanisms in DeFi lending protocols: A cross-chain comparative analysis of Aave and Compound," *arXiv preprint arXiv:2506.12855*, 2025.

[18] DeFiLlama, "DeFi lending market by TVL (april 2026 snapshot)," https://defillama.com/protocols/lending, 2026, accessed 2026-04-XX.

[19] T. Lim, "Predictive crypto-asset automated market maker architecture for decentralized finance using deep reinforcement learning," *Financial Innovation*, Vol. 10, No. 1, 2024.

[20] C.-L. Hwang and K. Yoon, *Multiple Attribute Decision Making: Methods and Applications*. Springer-Verlag, 1981.

[21] J.-P. Brans and P. Vincke, "A preference ranking organisation method: The PROMETHEE method for multiple criteria decision-making," *Management Science*, Vol. 31, No. 6, pp. 647–656, 1985.

[22] Z. Aljinović, B. Marasović, and T. Šestanović, "Cryptocurrency portfolio selection — a multicriteria approach," *Mathematics*, Vol. 9, No. 14, p. 1677, 2021.

[23] M. Hosseinzadeh *et al.*, "An asymmetric PROMETHEE II for cryptocurrency portfolio allocation based on return prediction," *Computers & Operations Research*, 2023.

[24] Anonymous, "AI-managed ERC-4626 yield vault with multi-criteria decision making," Anonymous preprint, 2026, anonymous prior work, citation withheld for blind review.

[25] K. Cho, B. van Merriënboer, C. Gulcehre, D. Bah-

danau, F. Bougares, H. Schwenk, and Y. Bengio, "Learning phrase representations using RNN encoder-decoder for statistical machine translation," *Proceedings of EMNLP*, 2014.

[26] K. Simonyan and A. Zisserman, "Two-stream convolutional networks for action recognition in videos," *Advances in Neural Information Processing Systems (NeurIPS)*, 2014.

[27] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.